# APACHE HADOOP MAPREDUCE

Online Shopping Order Analysis

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**Semester**: 7th

## Aim:

To write a MapReduce program in Python using the mrjob library that reads online shopping order data stored in
HDFS and calculates the total number of orders received for each product.

## Problem Statement:

An online shopping platform generates a large number of customer order records every day. Analysing these
records manually to determine the number of orders received for each product can be time-consuming when the
dataset becomes large. The order data is stored in the Hadoop Distributed File System (HDFS) and processed
using MapReduce.
In the Map phase, each order record is split into fields and the product name is extracted. The Mapper outputs
the pair (product, 1). Hadoop then performs Shuffle and Sort to group all values belonging to the same product.
In the Reduce phase, the values for each product are added to calculate the total number of orders.

**Tools and Environment Used:**

| | |
|---|---|
| **Operating System** | Ubuntu 22.04 LTS |
| **Java** | OpenJDK 11 |
| **Hadoop** | Apache Hadoop 3.4.3 |
| **Language** | Python 3.10 |
| **Library** | mrjob |
| **Storage** | HDFS |
| **Processing** | Hadoop MapReduce / YARN |
| **Setup** | Single-node Hadoop cluster |

**Input Data:**

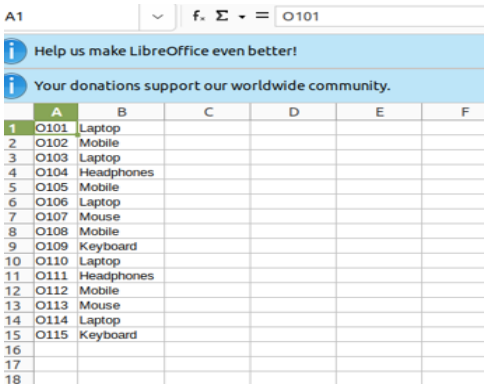
The input file orders.csv contains 15 online shopping order records. Each record contains an Order ID and

Product name. The file is stored in HDFS at /online_shopping/orders.csv.

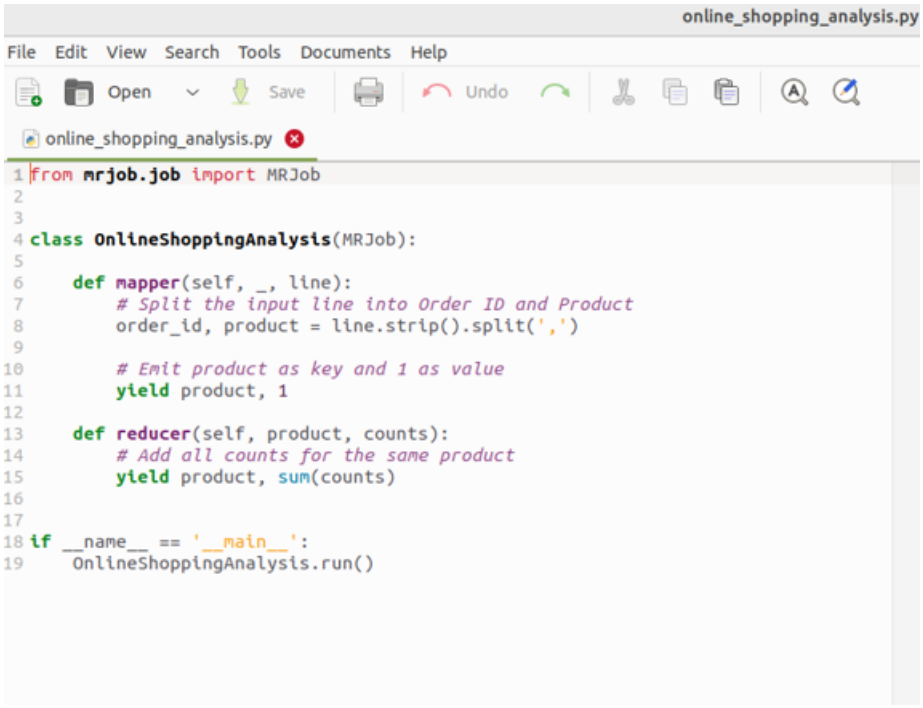
**Code Implementation:**

**File name:** online_shopping_analysis.py

**Dataset:**



| | A | B | C | D | E | F |
|---|---|---|---|---|---|---|
| 1 | O101 | Laptop | | | | |
| 2 | O102 | Mobile | | | | |
| 3 | O103 | Laptop | | | | |
| 4 | O104 | Headphones | | | | |
| 5 | O105 | Mobile | | | | |
| 6 | O106 | Laptop | | | | |
| 7 | O107 | Mouse | | | | |
| 8 | O108 | Mobile | | | | |
| 9 | O109 | Keyboard | | | | |
| 10 | O110 | Laptop | | | | |
| 11 | O111 | Headphones | | | | |
| 12 | O112 | Mobile | | | | |
| 13 | O113 | Mouse | | | | |
| 14 | O114 | Laptop | | | | |
| 15 | O115 | Keyboard | | | | |
| 16 | | | | | | |
| 17 | | | | | | |
| 18 | | | | | | |



```
from mrjob.job import MRJob


class OnlineShoppingAnalysis(MRJob):

    def mapper(self, _, line):
        # Split the input line into Order ID and Product
        order_id, product = line.strip().split(',')

        # Emit product as key and 1 as value
        yield product, 1

    def reducer(self, product, counts):
        # Add all counts for the same product
        yield product, sum(counts)


if __name__ == '__main__':
    OnlineShoppingAnalysis.run()
```

**mapper():** Splits each order record at the comma, extracts the product name, and outputs (product, 1).

**reducer():** Receives one product with all its counts and adds them using sum() to find the total number of orders.

**run():** Starts the mrjob program. The option -r hadoop submits the job to Hadoop using Hadoop Streaming.

**Execution Steps:**

1. Start Hadoop services and check that they are running:

**>>start-dfs.sh**

**>>start-yarn.sh**

**>>jps**

2. Activate the Python environment and check the input file:

**>>cd ~/online_shopping_analysis**

**>>source mrjob-env/bin/activate**

**>>wc -l orders.csv**

3. Upload the input file to HDFS:

**>>hdfs dfs -mkdir -p /online_shopping**

**>>hdfs dfs -put orders.csv /online_shopping/**

**>>hdfs dfs -ls /online_shopping**

4. Test the MapReduce program locally:

**>>python online_shopping_analysis.py orders.csv**

5. Run the MapReduce job on Hadoop:

**>>python online_shopping_analysis.py -r Hadoop**

**hdfs:///online_shopping/orders.csv**

6. Save and view the final output:

**>>python online_shopping_analysis.py -r hadoop hdfs :///online_shopping_orders.csv 2>/dev/null > output**

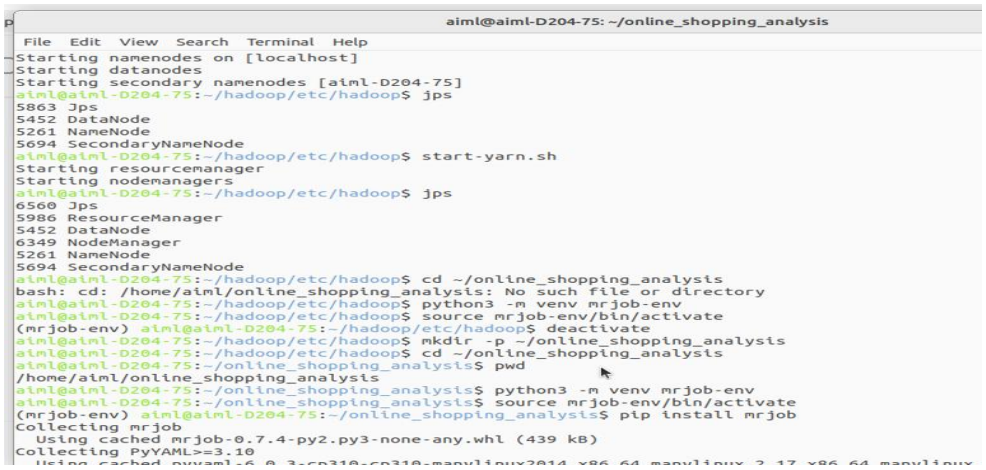



Fig:Starting HDFS and YARN – jps shows NameNode, DataNode, SecondaryNameNode, ResourceManager, NodeManager

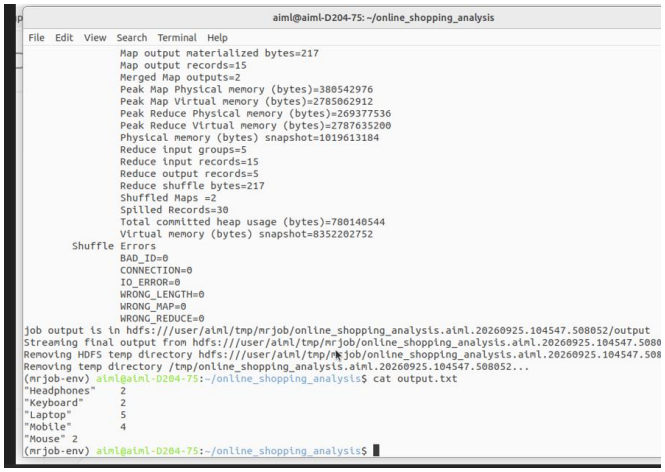






Fig:Job counters, output files (_SUCCESS and part-00000) in HDFS and the final result.

## Execution Output:

The Hadoop MapReduce job completed successfully. The final output was:

| Product | Number of Orders |
|---|---|
| Headphones | 2 |
| Keyboard | 2 |
| Laptop | 5 |
| Mobile | 4 |
| Mouse | 2 |

**Brief Discussion:**

The program works in three stages: **Map, Shuffle and Sort, and Reduce**. In the Map stage, each order record is read and the product name is extracted. The Mapper generates (product, 1) for every order. Hadoop then groups all values belonging to the same product during the Shuffle and Sort stage. In the Reduce stage, the values are added to calculate the total number of orders for each product.

| Product | Map Output Values | Reduce Output |
|---|---|---|
| Laptop | 1,1,1,1,1 | 5 |
| Mobile | 1,1,1,1 | 4 |
| HeadPhones | 1,1 | 2 |
| Mouse | 1,1 | 2 |
| Keyboard | 1,1 | 2 |

## Conclusion:

The Online Shopping Order Analysis MapReduce program was successfully implemented using Python and mrjob on a single-node Hadoop cluster running on Ubuntu. The order data was stored in HDFS and processed using Hadoop MapReduce. The Mapper extracted the product names and generated a count of 1 for each order, while the Reducer calculated the total number of orders for each product. The final results were obtained successfully, demonstrating how Hadoop can be used to process and analyse large amounts of data efficiently.